

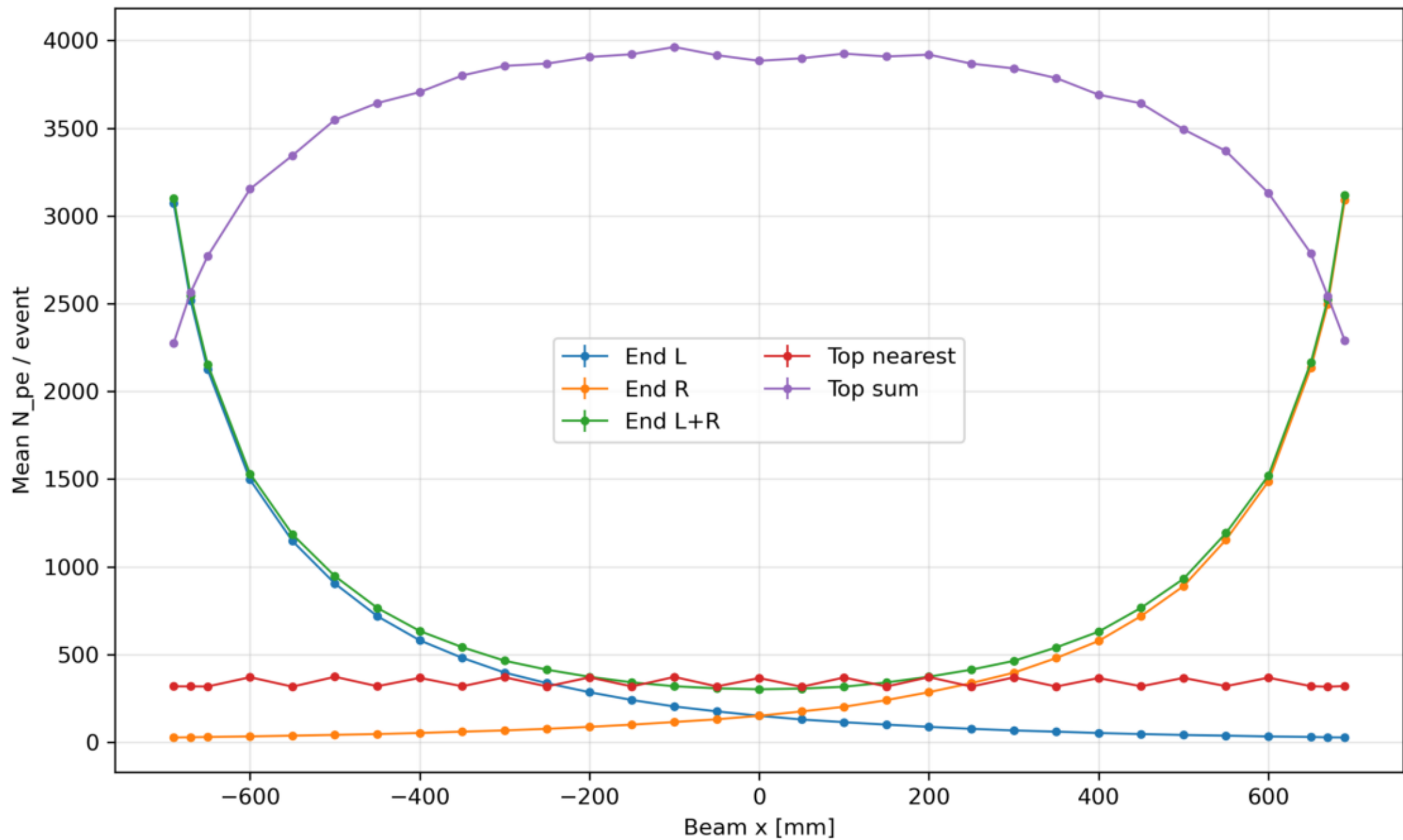
EXEC_08 - EXEC_07 EndTop photon budget

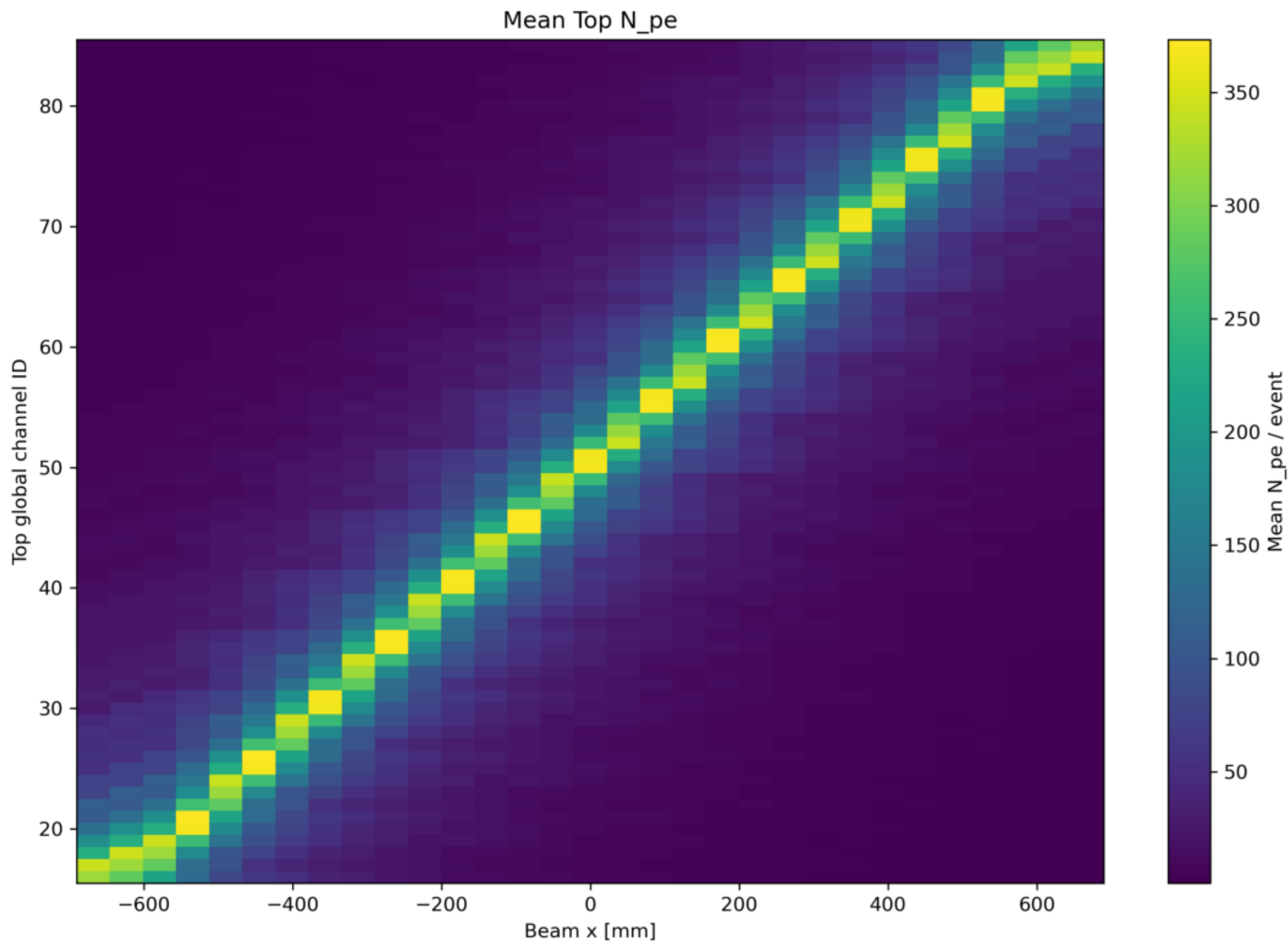
31 positions x 2000 events, 86 channels, EJ-230 OPSC-106, Broadcom AFBR-S4N66P024M

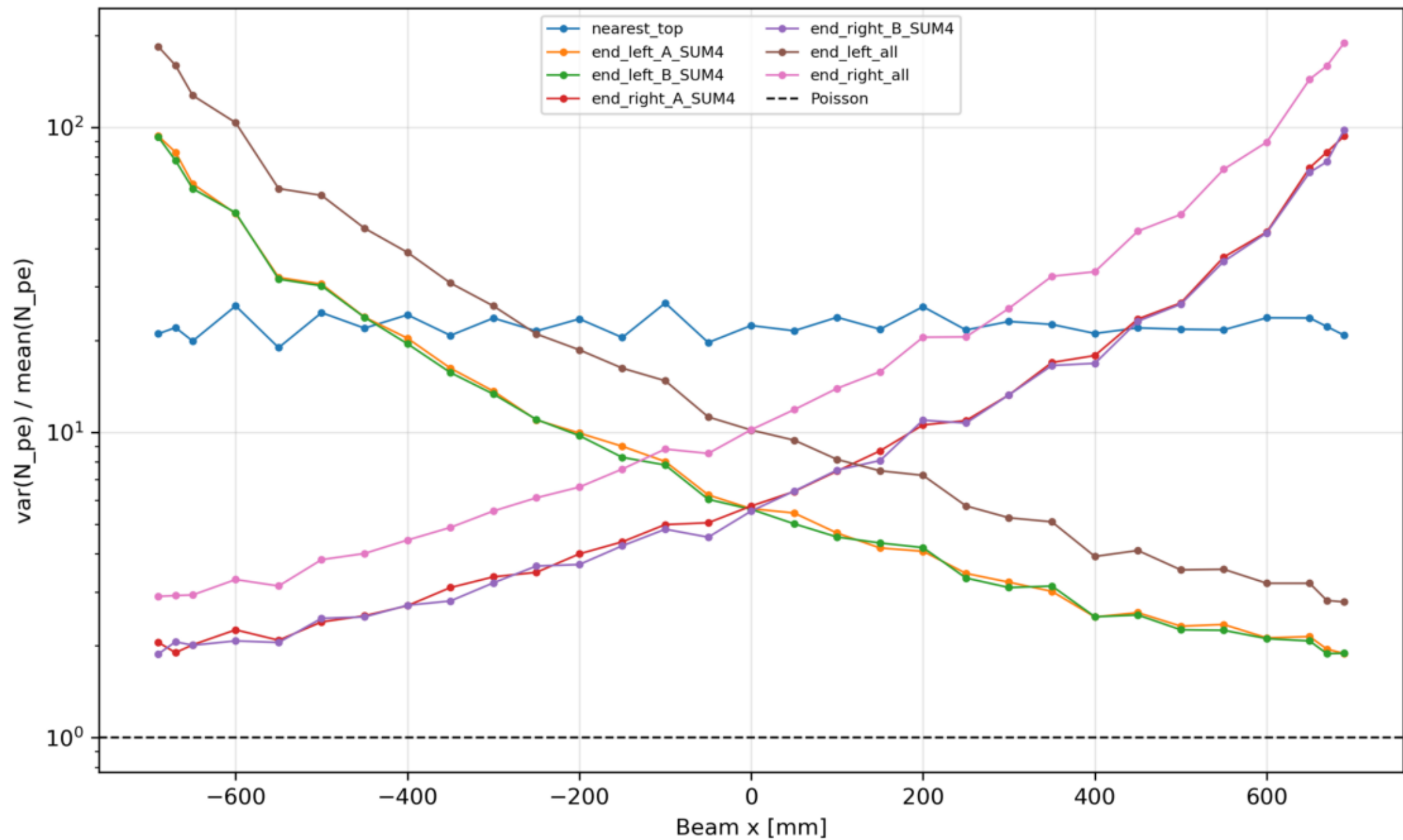
Intrinsic simulation: /sipm/jitterSigma 0

- Effective attenuation: left 30.70 +/- 0.02 cm; right 30.82 +/- 0.02 cm. These include light extraction through 70 Top windows and are not expected to equal
- Apparent slope-derived velocity from mean all-photon time: left 27.81 +/- 0.26 cm/ns; right 27.82 +/- 0.26 cm/ns. EXEC_10 shows this is not a direct propagation
- Typical nearest-Top var/mean: median 22.00, range 18.94-26.54. N_{pe} is not forced to Poisson.
- Intrinsic End SUM4 sigma_group= sigma(DeltaT)/sqrt(2): mean 176.71 ps, range 11.86-615.15 ps. No SPTR/electronics jitter.
- sigma(t_avg) mean 172.36 ps, range 93.69-325.87 ps.

Top uses 70 simulated SiPM elements and does not replicate the 32-SiPM hardware setup.

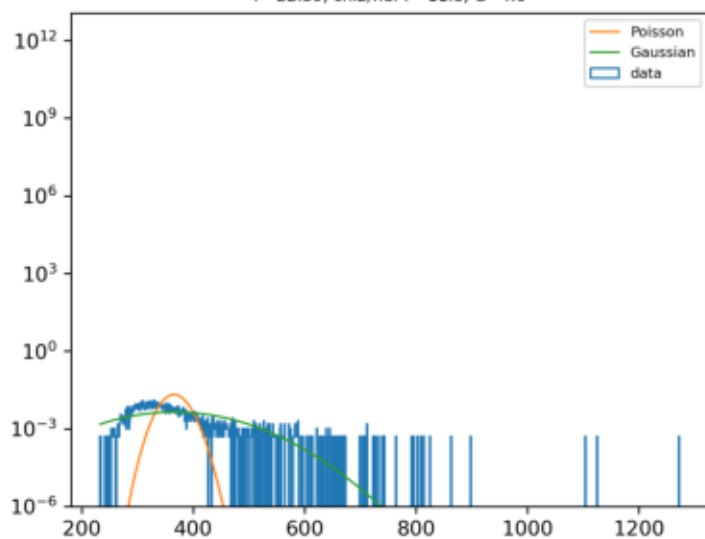




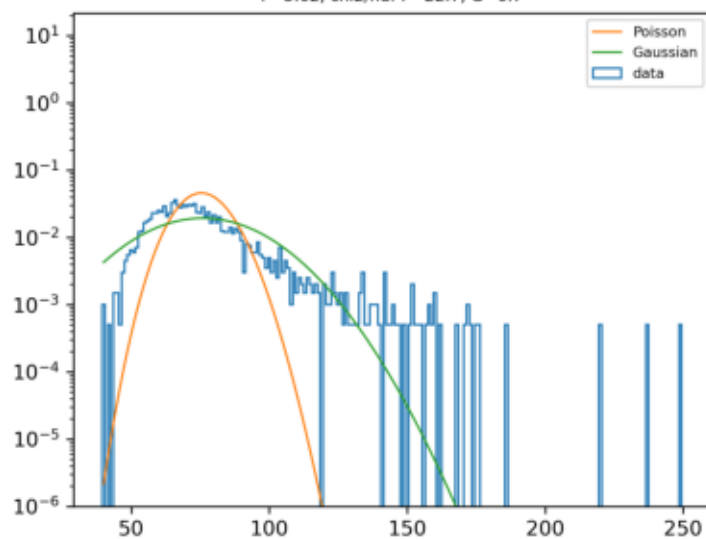


Poisson tail check, x=0 mm

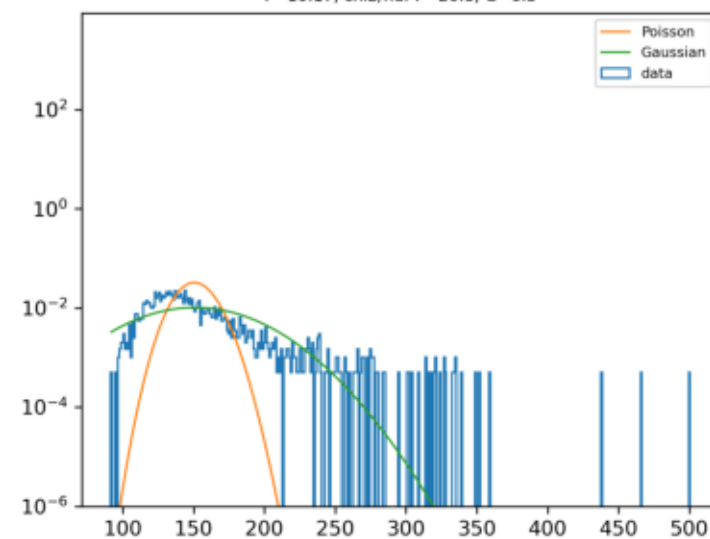
nearest_top
F=22.39, χ^2/ndf P=11.8, G=4.6

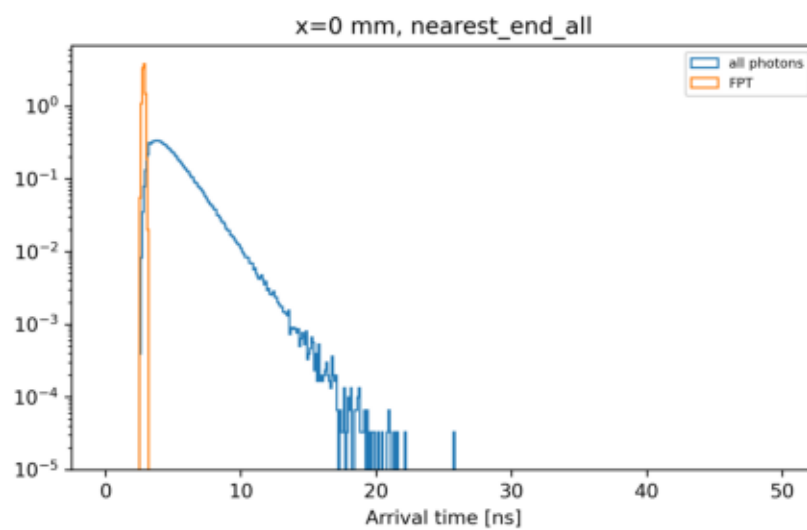
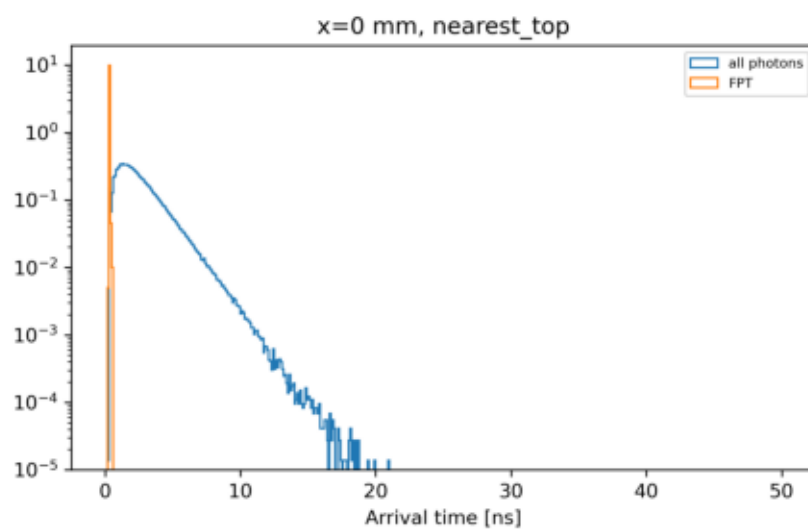
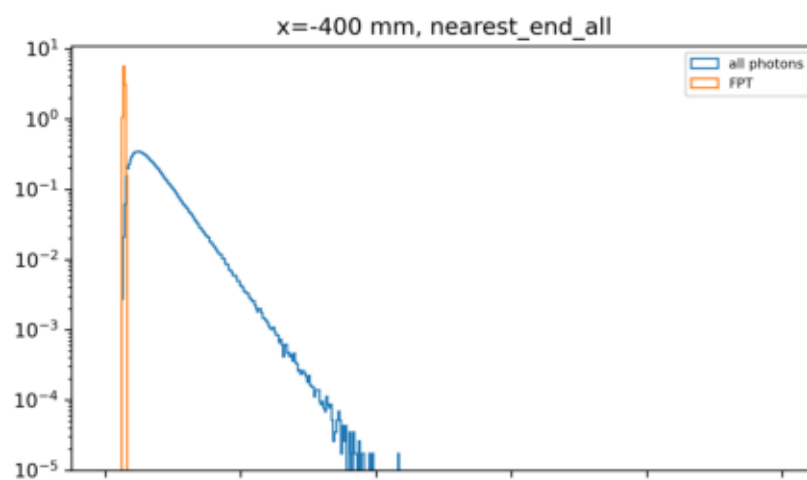
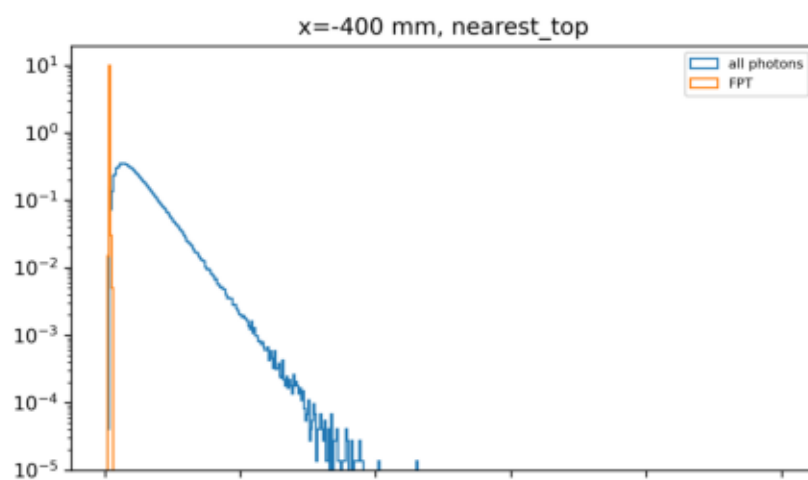
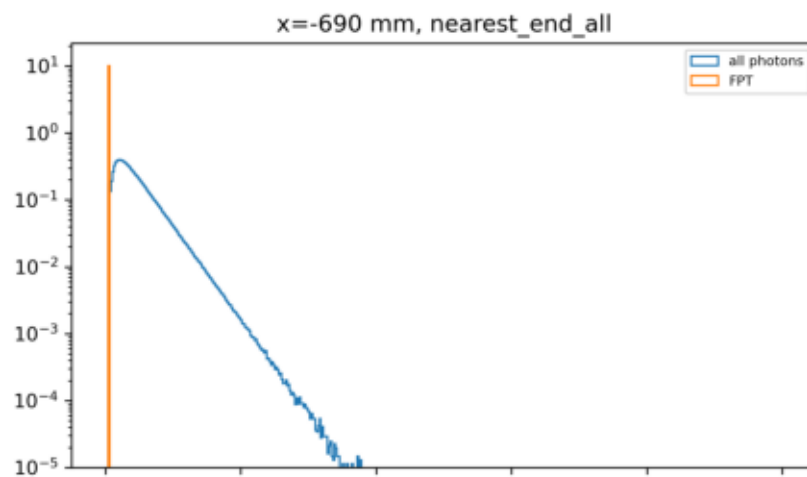
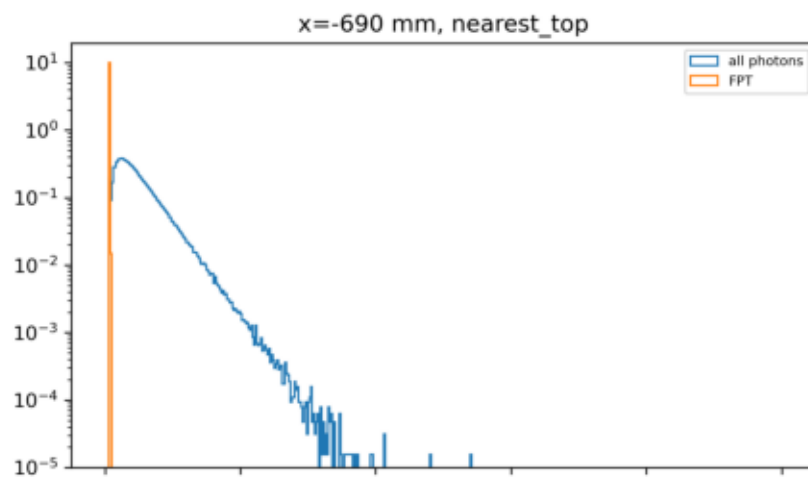


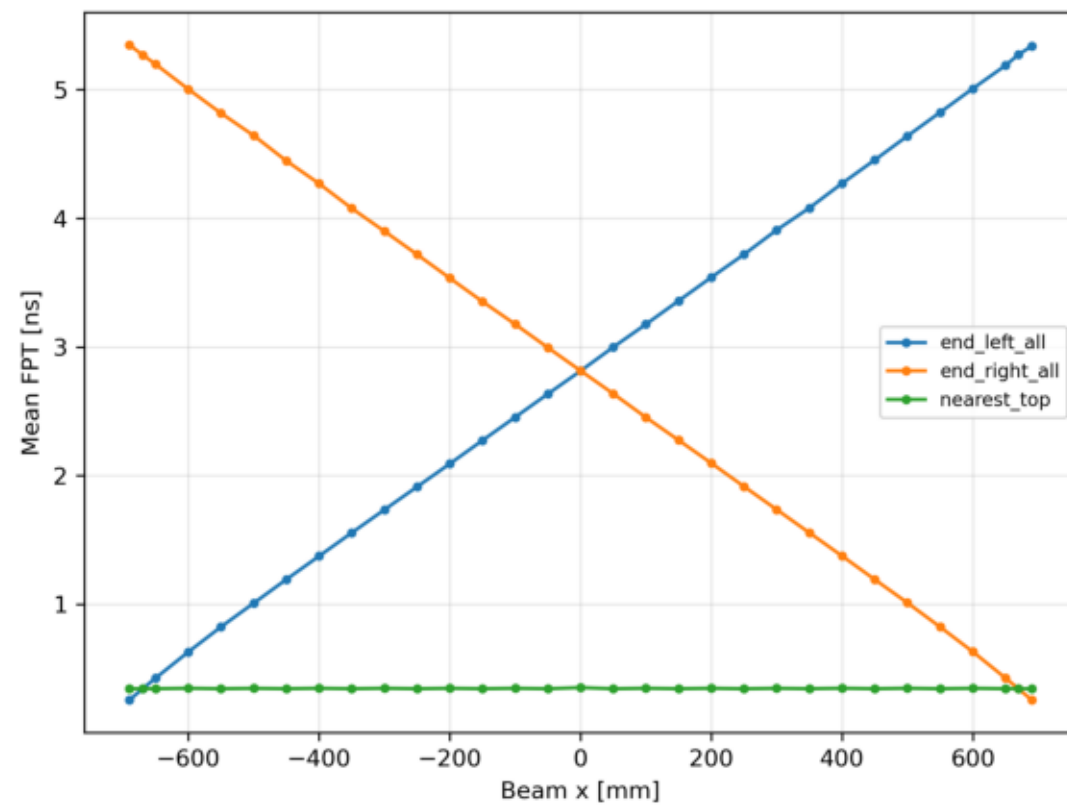
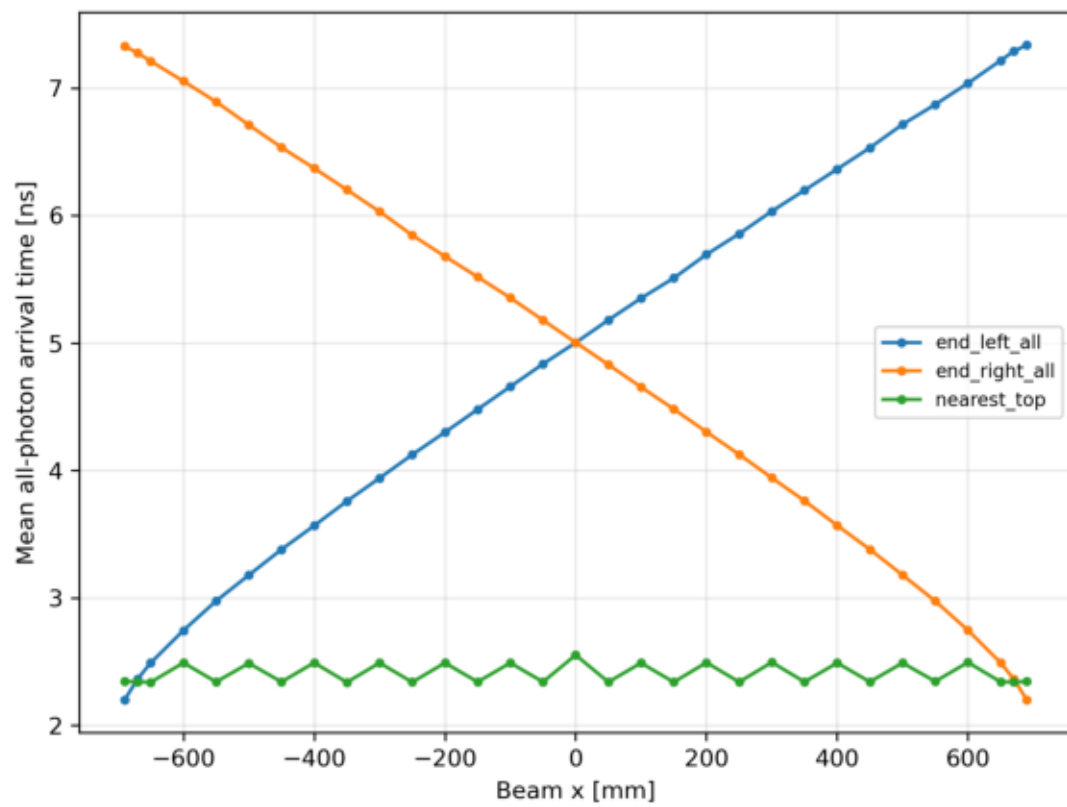
nearest_end_SUM4
F=5.62, χ^2/ndf P=22.7, G=6.7



nearest_end_all
F=10.17, χ^2/ndf P=20.8, G=6.3







Intrinsic End timing, no SPTR/electronics jitter

